AIRBNB

BERLIN · VIENNA · MUNICH



TEAM STRUDEL

01 · REGION AND CITIES

TWO CITIES TO TRAIN, ONE TO TEST

A · TRAINING

BERLIN

12,376 listings

snapshot 2026-09-19

B · TRAINING

VIENNA

12,982 listings

snapshot 2026-09-17

C · NEW CLIENT

MUNICH

7,493 listings

snapshot 2026-09-23

Why Munich. Same region and currency, but a much more expensive market. A model that only knows Berlin and Vienna prices should fail there, and we wanted to see it.

WHAT THE DATA TOLD US

34% and 24% of listings have no price

Not at random: 96–98% of listings closed for the whole year have no price. We do not fill it in; these rows stay in the data but are left out of training.

A long tail of prices

A typical price is about €130, but some listings ask over €9,000. So we predict log price and flag outliers per city and room type.

Prices are in EUR, not $

The dollar sign is an Inside Airbnb format artefact. The price-quote field confirms the currency.

The schema is not stable

Some columns are empty in both cities, some only in Vienna. Four Vienna district names arrive with broken characters.

02 · LEAKAGE REVIEW

COLUMNS THAT ALREADY KNOW THE PRICE

estimated_revenue_l365d = price × estimated_occupancy_l365d

Checked: the equation holds for 100% of priced listings in both cities

Excluded	Why
estimated_revenue_l365d	Computed from the price
price_quote_* (5 columns)	Hold the price itself, or exist only when a price exists
estimated_occupancy_l365d	Depends on the price and is unknown for a new listing
availability_30 / 60 / 90	Cheap listings get booked more: partly a result of the price
id, host_id, name, description	The model would memorise specific listings

MEDALLION LAKEHOUSE

SOURCE

Inside Airbnb

listings.csv.gz

Volume raw/ <city>/

BRONZE

bronze_listings

The raw file as text, 90 columns + city, snapshot, source, load time

SILVER

silver_listings

quarantine_listings

Cleaned, typed listings, the EDA rules, outlier flags

GOLD

gold_listing_features

gold_city_metrics

gold_neighbourhood_metrics

gold_city_centre

written by the model:

gold_price_predictions

gold_model_metrics

Price model

AI/BI dashboard

Genie Agent

Job airbnb_price_pipeline — **city** | snapshot_date | currency

bronze - silver - gold - checks, the same for berlin, vienna and munich. Every stage writes a line to pipeline_run_log

31 FEATURES WE CAN EXPLAIN

Size
guests, bedrooms, beds, bathrooms

Stay length
minimum nights, long stay

Location
distance to centre, city, coordinates

Demand and quality
reviews, scores, availability

Host
superhost, number of listings

Amenities
count and 8 flags: kitchen, washer…

TARGET
Log price, converted back with exp

MODEL CHOICE
5 candidates, cross-validation on train only

TEST SET
20% per city, 3,673 listings, used once

BASELINE
Median price for the same room type and guests

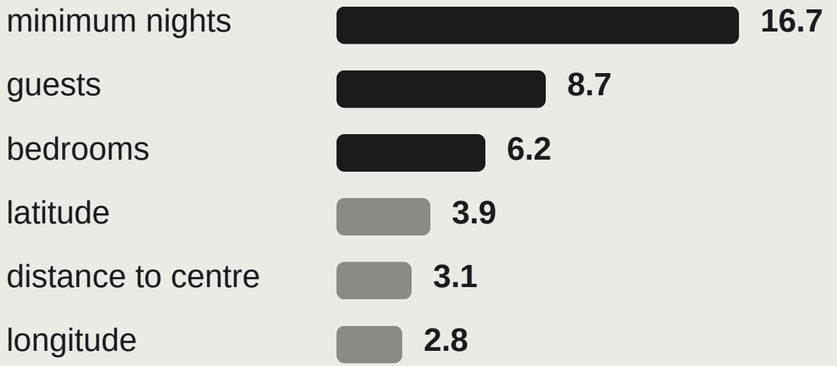
04 · TEST-SET RESULT

31% BETTER THAN THE BASELINE

MAE, EUR	Baseline	Model	Better
Berlin	63.80	42.85	32.8%
Vienna	54.85	38.33	30.1%
Both	58.93	40.39	31.5%

A typical listing is off by **about €22** (median error). A few expensive listings pull the MAE up to €40.

MAE increase without the feature, EUR



MUNICH: THE FIRST RUN

The pipeline ran

7,493
listings in bronze and silver

8 / 8
checks passed

0
quarantined, 0 duplicates

4
outliers flagged with no code change

The model failed

€150
MAE in Munich

€40
in Berlin and Vienna

Baseline in Munich: €163. Average error −€143: the model **under-prices almost every** listing.

TWO MODEL CHANGES

MODEL CHANGES

A remove city and coordinates
B price relative to the city median = v2
Trained on Berlin, Vienna only

Munich MAE, EUR, on the same 1,019 held-out listings



THREE NUMBERS SIDE BY SIDE

BASELINE IN MUNICH

€171

Berlin and Vienna medians

MODEL ON A AND B

€40

Berlin and Vienna test set

MODEL IN MUNICH

€157 → **€115**

v1 → v2, same 1,019 listings (v1 on all: €150)

Where the gap comes from

≈ €34

price level, the largest part

≈ €8

features that do not transfer

−€266

v2 still under-prices the expensive centre (Altstadt-Lehel)

The rest is Munich itself

Even Munich’s own baseline is twice as bad: €123 against €59 at home. Prices are higher and spread wider

06 · AI/BI DASHBOARD

EVERY TITLE IS A FINDING

1 · CITY OVERVIEW

“Munich is 60% more expensive than Berlin and Vienna, which cost almost the same”

Prices, availability, scores, hosts, room types, neighbourhoods

2 · BERLIN VS VIENNA

“Same price, but Berlin is a long-stay market and Vienna a professional short-stay one”

32% long stays against 11%; 2.31 listings per host in Vienna against 1.56

3 · MODEL

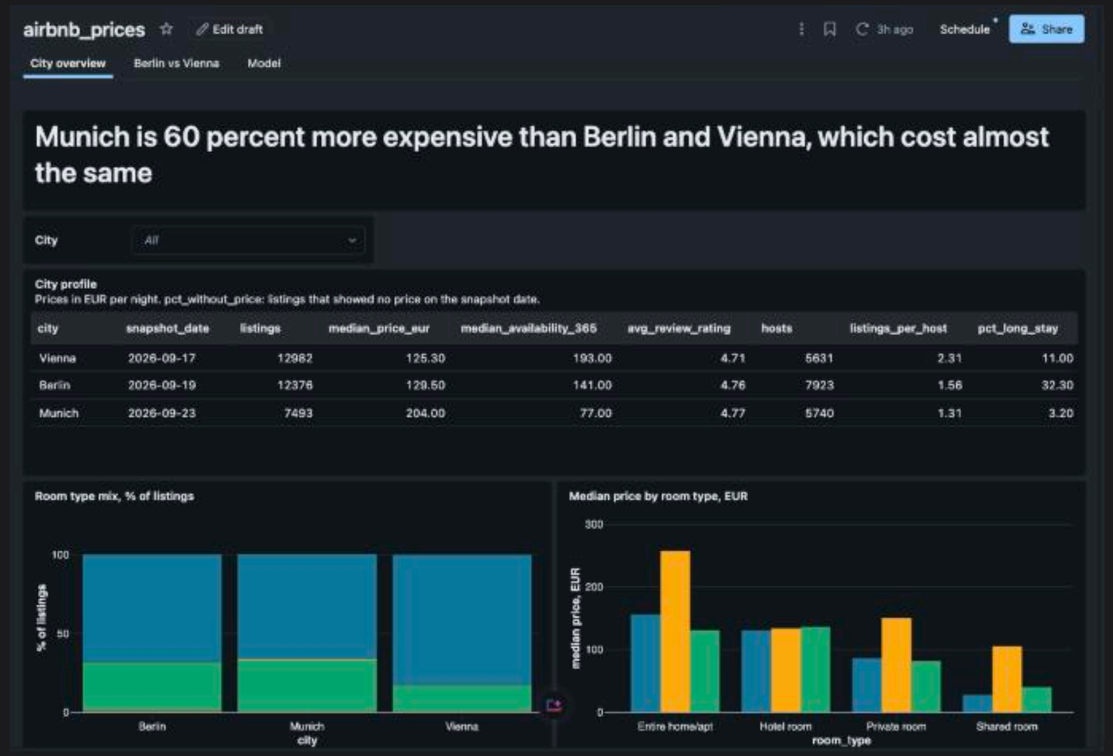
“31% better than the baseline at home; in Munich v2 cuts the error from €157 to €115”

v1 vs v2 on the same Munich listings, by district; errors by room type, top 20

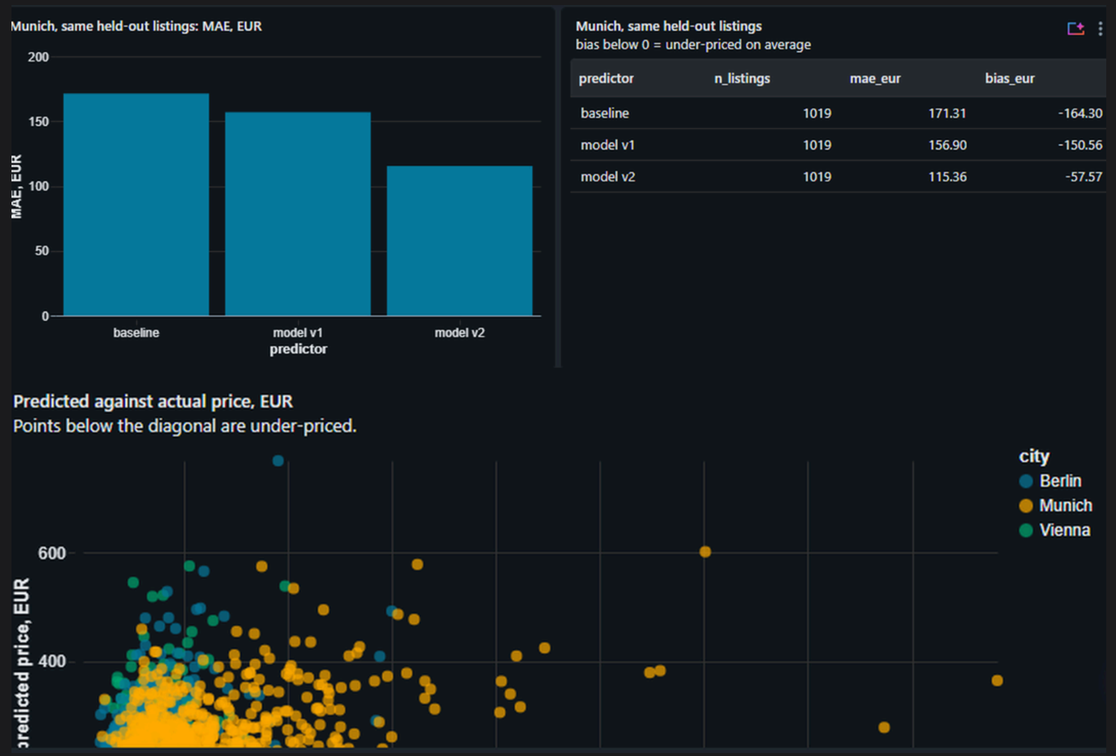
Honest errors: only on listings the model did not train on: the Berlin and Vienna test set; in Munich all listings for v1, held-out ones for v2.

Version as a filter: v2 appeared by itself; v1 and v2 are compared on the same 1,019 Munich listings.

CITY OVERVIEW AND MODEL



Page 1 · city profile: price, availability, hosts, long stays



Page 3 · Munich: baseline, v1 and v2 on the same 1,019 listings

HOW WE GOT GENIE TO ANSWER CORRECTLY

Median price of entire homes in Berlin?

€155.40

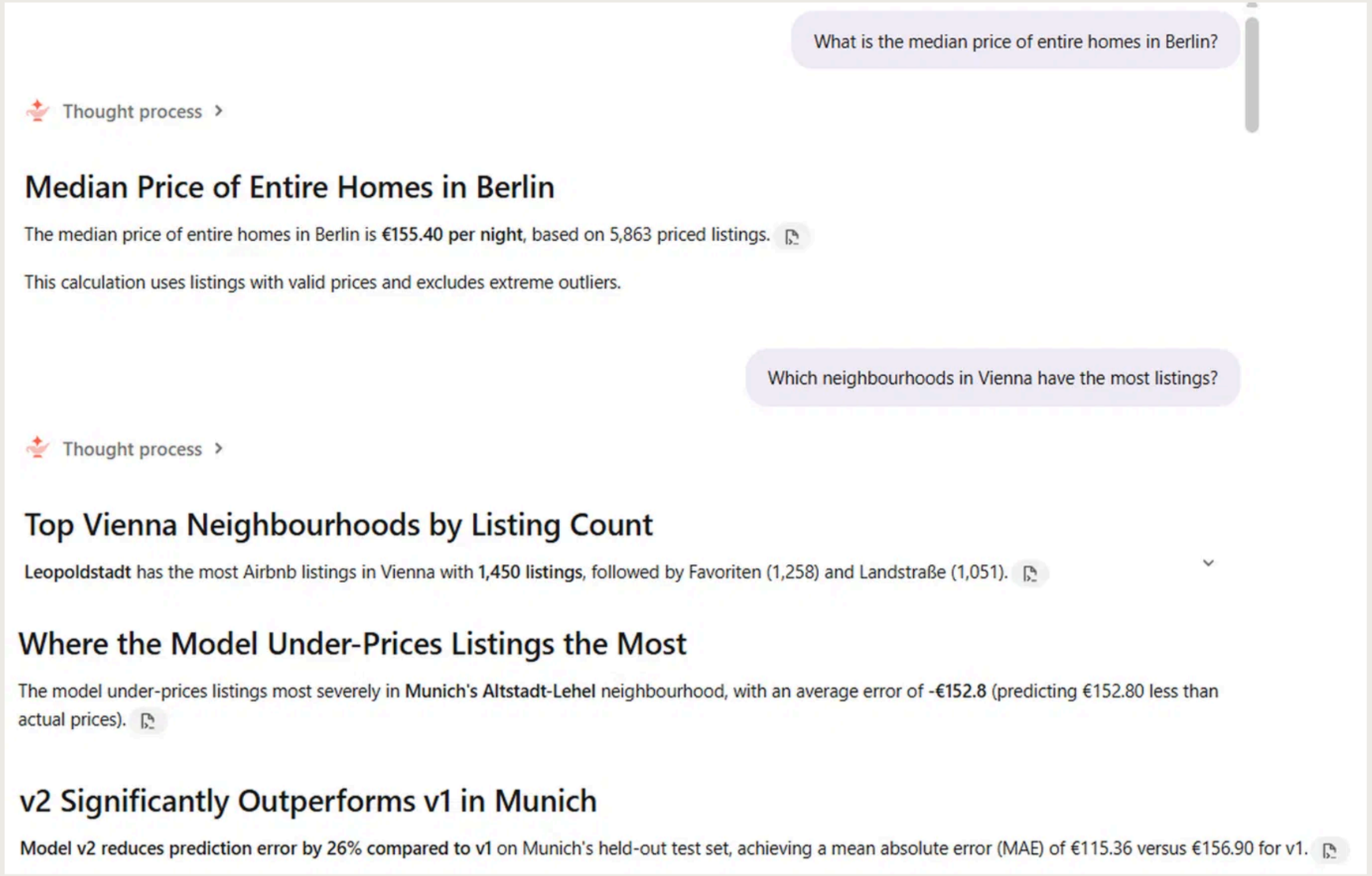
Where in Vienna are the most listings?

Leopoldstadt (1,450), then Favoriten and Landstraße

Where does the model under-price most?

Munich, Altstadt-Lehel

1. The pipeline writes table and column descriptions, and Genie reads them
2. One short rule per mistake: lower-case cities, one model version, the sign of the error, same listings for v1 vs v2
3. Trusted example SQL for the three key questions
4. Value sampling: “Wien” and “entire homes” map to the stored values
5. Sample questions on the start screen


What is the median price of entire homes in Berlin?
Thought process
Median Price of Entire Homes in Berlin
The median price of entire homes in Berlin is €155.40 per night, based on 5,863 priced listings.
This calculation uses listings with valid prices and excludes extreme outliers.
Which neighbourhoods in Vienna have the most listings?
Thought process
Top Vienna Neighbourhoods by Listing Count
Leopoldstadt has the most Airbnb listings in Vienna with 1,450 listings, followed by Favoriten (1,258) and Landstraße (1,051).
Where the Model Under-Prices Listings the Most
The model under-prices listings most severely in Munich's Altstadt-Lehel neighbourhood, with an average error of -€152.8 (predicting €152.80 less than actual prices).
v2 Significantly Outperforms v1 in Munich
Model v2 reduces prediction error by 26% compared to v1 on Munich's held-out test set, achieving a mean absolute error (MAE) of €115.36 versus €156.90 for v1.

CHALLENGES

Line breaks in the CSV

Host texts broke the read. Fix: multiLine with quote and escape options

ANSI mode on serverless

Plain casts failed on dirty values. Fix: try_cast everywhere

Vienna encoding

District names read with the wrong code page. Found in the EDA, fixed in silver

Festival peak

Oktoberfest cannot be checked with one snapshot

07 · TEAM

WHO DID WHAT

01

Bozhena

Data and analysis

Data loading, EDA, leakage review, cleaning rules, architecture, presentation

02

Yustyna Haievska

Pipeline

Parameterised Job, bronze, silver, gold and checks, Munich first run

03

Anna Kryvulia

Model

Features, model choice, test-set evaluation, predictions and metrics in gold, model v2, the gap explained

04

Oksana Dziuba

Baseline, dashboard, Genie

Baseline and its Spark check, AI/BI dashboard, Genie Agent, repository

THANK YOU

